

8.3 Nonhomogeneous Linear Systems

Consider the nonhomogeneous system:

$$\vec{X}'(t) = A \vec{X}(t) + \vec{F}(t) \quad [NH]$$

Where A is an $n \times n$ matrix and $\vec{X}(t)$ and $\vec{F}(t)$ are column vectors in $\mathbb{R}^n$.

The general solution of $[NH]$ has the

form:
$$\vec{X}(t) = \vec{X}_c(t) + \vec{X}_p(t)$$

where:

$\vec{X}_c(t)$ is the general solution of $\vec{X}'(t) = A \vec{X}(t)$

and

$\vec{X}_p(t)$ is a particular solution of $[NH]$.

In this section, we present the method of Variation of parameters to find $\vec{X}_p(t)$.

Assume that we have n linearly independent solutions of $\vec{X}'(t) = A \vec{X}(t)$

$\vec{X}_1(t), \vec{X}_2(t), \dots, \vec{X}_n(t)$ so that

$$\vec{X}_c(t) = c_1 \vec{X}_1(t) + c_2 \vec{X}_2(t) + \dots + c_n \vec{X}_n(t).$$

$$\Phi(t) = \left[\vec{X}_1(t) \mid \vec{X}_2(t) \mid \dots \mid \vec{X}_n(t) \right]$$

is a Fundamental matrix of the

system $\vec{X}'(t) = A \vec{X}(t)$.

Properties of Φ

$$\textcircled{1} \quad \Phi'(t) = A \Phi(t).$$

Since each column of $\Phi(t)$ is a solution of $\vec{X}'(t) = A \vec{X}(t)$.

$$\textcircled{2} \quad \Phi(t) \text{ is invertible.}$$

Since the columns of $\Phi(t)$ are L.I.

To find a particular solution of $[NH]$,
we look for a solution in the form

$$\vec{X}_p(t) = \Phi(t) \vec{U}(t)$$

for some vector function $\vec{U}(t)$.

$$\vec{X}_p(t) = \Phi(t) \vec{U}(t)$$

$$\Rightarrow \vec{X}'_p(t) = \underbrace{\Phi'(t) \vec{U}(t) + \Phi(t) \vec{U}'(t)}_{\text{product rule}}$$

Assume that $\vec{X}_p(t)$ is a solution of

$$\vec{X}'(t) = A \vec{X}(t) + \vec{F}(t).$$

$$\Rightarrow \Phi'(t) \vec{U}(t) + \Phi(t) \vec{U}'(t) = A \Phi(t) \vec{U}(t) + \vec{F}(t)$$

Since $\Phi'(t) = A \Phi(t)$, then:

$$\cancel{A \Phi(t) \vec{U}(t)} + \Phi(t) \vec{U}'(t) = \cancel{A \Phi(t) \vec{U}(t)} + \vec{F}(t)$$

$$\Rightarrow \Phi(t) \vec{U}'(t) = \vec{F}(t)$$

Since $\Phi(t)$ is invertible, then:

$$\vec{U}'(t) = \Phi^{-1}(t) \vec{F}(t)$$

$$\Rightarrow \vec{U}(t) = \int \Phi^{-1}(t) \vec{F}(t) dt$$

since $\vec{X}_p(t) = \Phi(t) \vec{U}(t)$

then:

$$\vec{X}_p(t) = \Phi(t) \int \Phi^{-1}(t) \vec{F}(t) dt$$

The general solution of

$$\vec{X}'(t) = A \vec{X}(t) + \vec{F}(t)$$

is

$$\vec{X}(t) = \vec{X}_c(t) + \vec{X}_p(t)$$

where:

$$\vec{X}_c(t) = c_1 \vec{X}_1(t) + c_2 \vec{X}_2(t) + \dots + c_n \vec{X}_n(t)$$

$$= \Phi(t) \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix}$$

and

$$\vec{X}_p(t) = \Phi(t) \int \Phi^{-1}(t) \vec{F}(t) dt.$$

You don't need to include a constant of integration when you find $\vec{X}_p(t)$ since any constant of integration would be absorbed by $\vec{X}_c(t)$.

ex: Find the solution of

$$\begin{cases} x'(t) = 2x(t) - y(t) \\ y'(t) = 3x(t) - 2y(t) + 4t \end{cases}$$

With $x(0) = 5$ and $y(0) = 7$.

$$\begin{bmatrix} x(t) \\ y(t) \end{bmatrix}' = \underbrace{\begin{bmatrix} 2 & -1 \\ 3 & -2 \end{bmatrix}}_A \underbrace{\begin{bmatrix} x(t) \\ y(t) \end{bmatrix}}_{\vec{X}(t)} + \underbrace{\begin{bmatrix} 0 \\ 4t \end{bmatrix}}_{\vec{F}(t)}$$

$$\vec{X}(0) = \begin{bmatrix} 5 \\ 7 \end{bmatrix}$$

Step ① Find $\vec{X}_c(t)$

We need to find the eigenvalues and eigenvectors of A .

$$A = \begin{bmatrix} 2 & -1 \\ 3 & -2 \end{bmatrix}$$

Eigenvalue

$$\lambda_1 = 1$$

$$\lambda_2 = -1$$

Eigenvector

$$\vec{K}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\vec{K}_2 = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

$$\vec{X}_c(t) = c_1 \vec{K}_1 e^{\lambda_1 t} + c_2 \vec{K}_2 e^{\lambda_2 t}$$

$$\vec{X}_c(t) = c_1 \underbrace{\begin{bmatrix} 1 \\ 1 \end{bmatrix} e^t}_{\vec{X}_1(t)} + c_2 \underbrace{\begin{bmatrix} 1 \\ 3 \end{bmatrix} e^{-t}}_{\vec{X}_2(t)}$$

Step ② Find $\vec{X}_p(t)$

$$\left. \begin{aligned} \vec{X}_1(t) &= \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^t = \begin{bmatrix} e^t \\ e^t \end{bmatrix} \\ \vec{X}_2(t) &= \begin{bmatrix} 1 \\ 3 \end{bmatrix} e^{-t} = \begin{bmatrix} e^{-t} \\ 3e^{-t} \end{bmatrix} \end{aligned} \right\} \text{From } \vec{X}_c(t)$$

$$\begin{aligned} \Phi(t) &= \left[\vec{X}_1(t) \mid \vec{X}_2(t) \right] \\ &= \begin{bmatrix} e^t & e^{-t} \\ e^t & 3e^{-t} \end{bmatrix} \end{aligned}$$

$$\vec{F}(t) = \begin{bmatrix} 0 \\ 4t \end{bmatrix}$$

$$\vec{X}_p(t) = \Phi(t) \int \Phi^{-1}(t) \vec{F}(t) dt$$

We need to find $\Phi^{-1}(t)$.

Recall : If $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, then

$$A^{-1} = \frac{1}{\det(A)} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

$$\det(\Phi(t)) = 3e^{-t}e^t - e^{-t}e^t = 2$$

$$\Rightarrow \Phi^{-1}(t) = \frac{1}{2} \begin{bmatrix} 3e^{-t} & -e^{-t} \\ -e^t & e^t \end{bmatrix}$$

$$\begin{aligned} \Phi^{-1}(t) \vec{F}(t) &= \frac{1}{2} \begin{bmatrix} 3e^{-t} & -e^{-t} \\ -e^t & e^t \end{bmatrix} \begin{bmatrix} 0 \\ 4t \end{bmatrix} \\ &= \begin{bmatrix} -2te^{-t} \\ 2te^t \end{bmatrix} \end{aligned}$$

$$\int \Phi^{-1}(t) \vec{F}(t) dt = \int \begin{bmatrix} -2te^{-t} \\ 2te^t \end{bmatrix} dt$$

We need an integration by parts for each component.

D		I
$-2t$	$\oplus$	e^{-t}
-2	$\ominus$	$-e^{-t}$
0		e^{-t}

D		I
$2t$	$\oplus$	e^t
2	$\ominus$	e^t
0		e^t

$$\int \Phi^{-1}(t) \vec{F}(t) dt = \begin{bmatrix} ate^{-t} + 2e^{-t} \\ ate^t - 2e^t \end{bmatrix}$$

$$\vec{X}_p(t) = \Phi(t) \int \Phi^{-1}(t) \vec{F}(t) dt$$

$$\vec{X}_p(t) = \begin{bmatrix} e^t & e^{-t} \\ e^t & 3e^{-t} \end{bmatrix} \begin{bmatrix} 2te^{-t} + 2e^{-t} \\ 2te^t - 2e^t \end{bmatrix}$$

$$\vec{X}_p(t) = \begin{bmatrix} (2t+2) + (2t-2) \\ (2t+2) + 3(2t-2) \end{bmatrix}$$

$$\vec{X}_p(t) = \begin{bmatrix} 4t \\ 8t-4 \end{bmatrix}$$

The general solution is

$$\vec{X}(t) = \vec{X}_c(t) + \vec{X}_p(t)$$

$$\vec{X}(t) = c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^t + c_2 \begin{bmatrix} 1 \\ 3 \end{bmatrix} e^{-t} + \begin{bmatrix} 4t \\ 8t-4 \end{bmatrix}.$$

Step ③ Find c_1 and c_2 .

$$\vec{X}(0) = \begin{bmatrix} 5 \\ 7 \end{bmatrix}$$

$$\Rightarrow c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ 3 \end{bmatrix} + \begin{bmatrix} 0 \\ -4 \end{bmatrix} = \begin{bmatrix} 5 \\ 7 \end{bmatrix}$$

$$\Rightarrow c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ 3 \end{bmatrix} = \begin{bmatrix} 5 \\ 11 \end{bmatrix}$$

$$\left[\begin{array}{cc|c} 1 & 1 & 5 \\ 1 & 3 & 11 \end{array} \right] \xrightarrow{\text{RREF}} \left[\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & 3 \end{array} \right] \Rightarrow \begin{cases} c_1 = 2 \\ c_2 = 3 \end{cases}$$

The solution of the IVP system is :

$$\vec{X}(t) = 2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^t + 3 \begin{bmatrix} 1 \\ 3 \end{bmatrix} e^{-t} + \begin{bmatrix} 4t \\ 8t-4 \end{bmatrix}$$

Since $\vec{X}(t) = \begin{bmatrix} x(t) \\ y(t) \end{bmatrix}$

then:

$$x(t) = 2e^t + 3e^{-t} + 4t$$

$$y(t) = 2e^t + 9e^{-t} + 8t - 4$$

The End!